PSG Institute of Technology and Applied Research

Department of Artificial Intelligence and Data Science

A Multi-Agent Reinforcement Learning Architecture with Independent Sentiment-Anomaly Veto for Pump-and-Dump-Resistant Trading

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Problem Statement

- Multi-agent RL trading systems increasingly use sentiment signals to react to market mood — but treat sentiment as one input among equals.
  - Fused into a shared state vector feeding an RL policy, or
  - Cast as one vote among several in a majority-vote decision layer
- This means a sufficiently coordinated manipulation (a pump-and-dump) can still succeed if it also moves price/volume signals in its favor — sentiment has no authority independent of the profit-seeking objective manipulation exploits.
- **Confirmed directly in our base papers:**
  - Singh's Agentic AI framework — sentiment is 1 of 3 votes in a majority-vote Decision Agent; can be outvoted 2-to-1 during a real manipulation event.
  - MSPM — sentiment is fused into the DQN's state vector as just another feature; can be down-weighted by the network during training.
- **Gap: no existing multi-agent trading architecture gives a sentiment-anomaly detector independent authority to override trade decisions during a detected manipulation event.**

Abstract

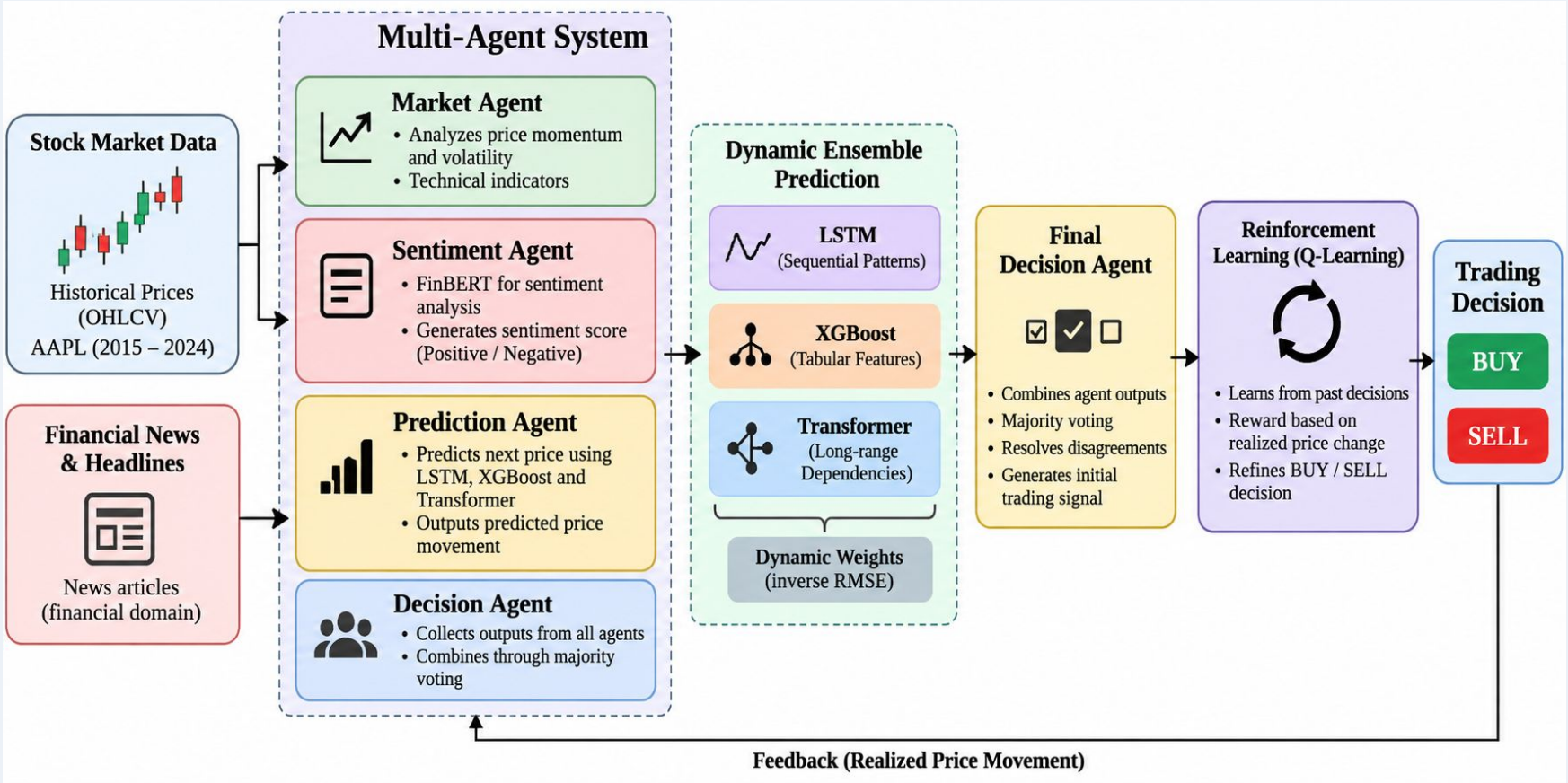
- Existing multi-agent reinforcement learning (MARL) trading systems incorporate sentiment as a signal fused into a shared decision space — as one vote among equals, or as one feature in a shared state vector — leaving them vulnerable to coordinated pump-and-dump manipulation that moves price and sentiment signals together.
- This project proposes a MARL trading architecture in which an Analysis Agent (price-action policy) and a Risk Agent (position/volatility limits) propose trades as usual, while an independently-trained Sentiment-Anomaly Agent scores real pump-and-dump coordination text against a rolling per-asset baseline to compute a Sentiment-Anomaly Index.
- When this index crosses a threshold, the agent exercises veto authority — forcing a hedge or reversal regardless of what the trading agents conclude — rather than contributing one more vote or feature that can be outvoted or down-weighted.
- The system is built on and evaluated against two base papers (an Agentic AI multi-agent stock-prediction framework, and MSPM's modular MARL portfolio system), using real labelled pump-and-dump datasets (La Morgia et al.; Ardia & Bluteau) rather than synthetic manipulation events, with an ablation study comparing veto-enabled vs. veto-disabled configurations.

BASE PAPER

Agentic AI for Intelligent Stock Market Prediction : A Multi-Agent Framework Combining LSTM, Transformer, XGBoost, FinBERT & Reinforcement Learning

Khushvir Singh, June 11, 2026

System Architecture



Discussions

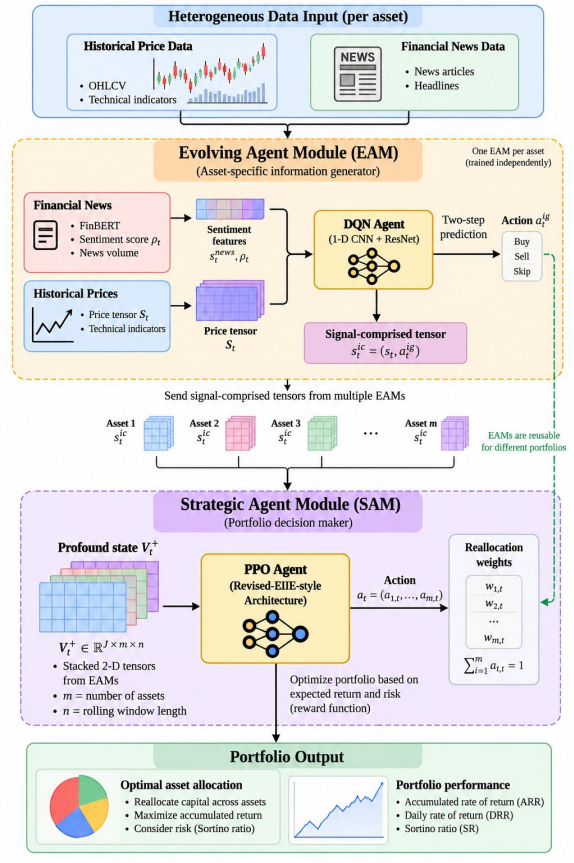
- **LSTM Performance:** LSTM is the core prediction model, achieving $R^2 = 0.915$ and MAE = $4.64.
- **Ensemble Learning:** Combining LSTM, XGBoost, and Transformer improves prediction performance, with the ensemble achieving approximately $R^2 = 0.96$.
- **Multi-Agent Coordination:** Market, Sentiment, Prediction, and Decision Agents combine different signals to generate trading decisions.
- **Risk Management:** Backtesting achieved 8.69% CAGR and −10.04% maximum drawdown; however, passive buy-and-hold AAPL produced higher raw returns.
- **Identified Limitations:** The Transformer was undertrained, randomized sentiment signals affected reinforcement learning, and transaction costs were not considered.
- **Future Improvements:** Integrate real-time financial news, train the Transformer longer, improve reinforcement learning, and evaluate multiple stocks with transaction costs.

Conclusion: The paper demonstrates the potential of combining multi-agent AI, ensemble learning, financial sentiment analysis, and reinforcement learning for stock-trading decisions.

LITERATURE SURVEY

MSPM : A Modularized & Scalable Multi-Agent Reinforcement Learning Based System for Financial Portfolio Management

Zhenhan Huang, Fumihide Tanaka



Key highlights

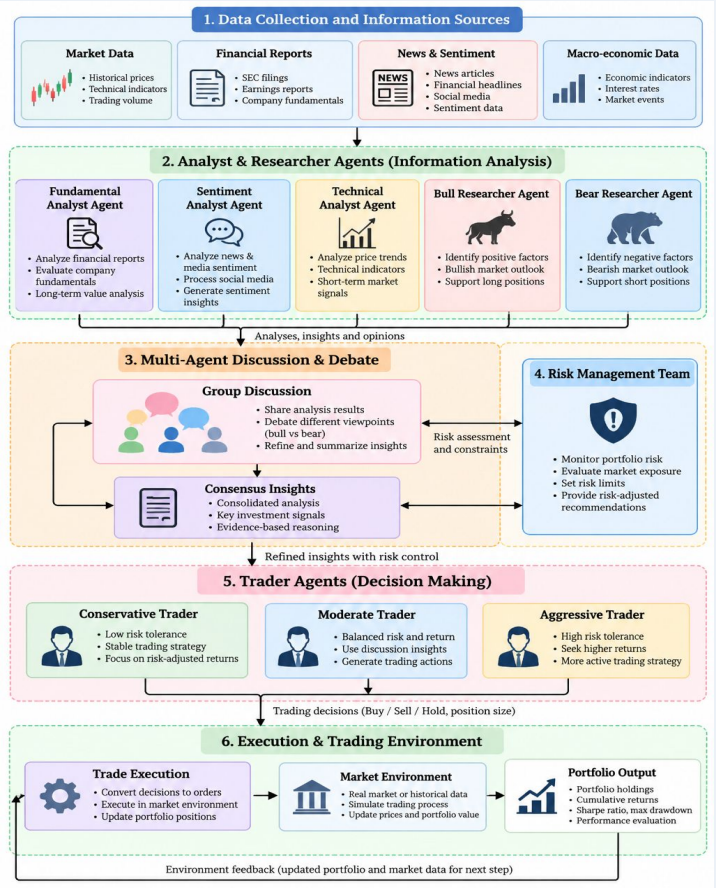
- **Modular Multi-Agent Framework:** Combines Evolving Agent Modules (EAM) and a Strategic Agent Module (SAM).
- **Market Analysis:** EAM uses DQN to process historical prices, technical indicators, and financial news sentiment.
- **Portfolio Optimization:** SAM uses PPO to determine asset allocation weights.
- **Scalability:** EAMs can be reused across different portfolios.
- **Performance:** Outperforms five baseline methods on ARR, DRR, and Sortino ratio.

Limitations

- Evaluates only DQN and PPO algorithms.
- Reward-function design involves trade-offs between risk stability and portfolio performance.
- Additional data sources, such as satellite imagery, remain future research directions.

Trading Agents : Multi - Agents LLM Financial Trading Framework

Yijia Xio et. al.



Key highlights

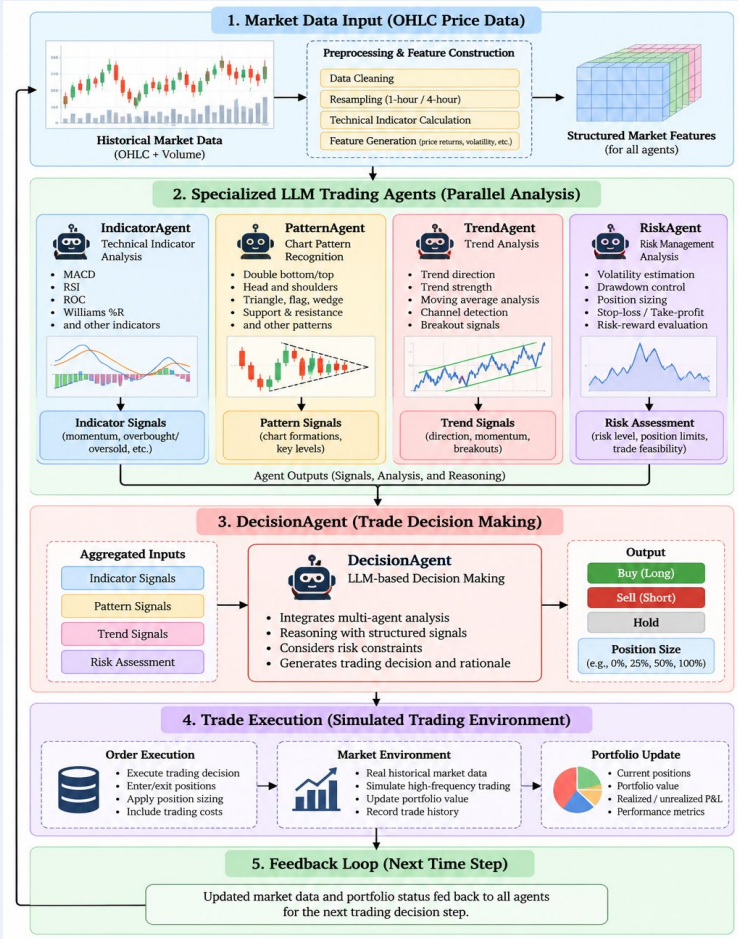
- **Multi-Agent LLM Framework**: Simulates a real-world trading firm using specialized AI agents.
- **Multi-Source Market Analysis**: Integrates historical stock prices, financial reports, technical indicators, news, and social media sentiment.
- **Bull–Bear Debate:** Bullish and bearish researcher agents debate market conditions to generate balanced investment insights.
- **Risk Management:** Risk-seeking, neutral, and conservative agents evaluate trading risks and refine decisions.
- **Performance & Explainability:** Outperforms five baseline strategies in key experiments, including cumulative returns and Sharpe ratio, while providing interpretable, evidence-based trading decisions.

Limitations

- **Short Evaluation Period**: Experiments cover only three months (January–March 2024).
- **High Computational Cost:** Extensive LLM and tool calls increase API usage and limit longer backtests.
- **Limited Real-World Validation:** Evaluation is based on historical backtesting; live-market deployment remains future work.
- **Dependence on LLMs and Data:** Decision quality depends on model reasoning and the availability and reliability of financial data.

QuantHarness: Price-Driven Multi-Agent LLMs for High-Frequency Trading

Fei Xiong et. al.



Key highlights

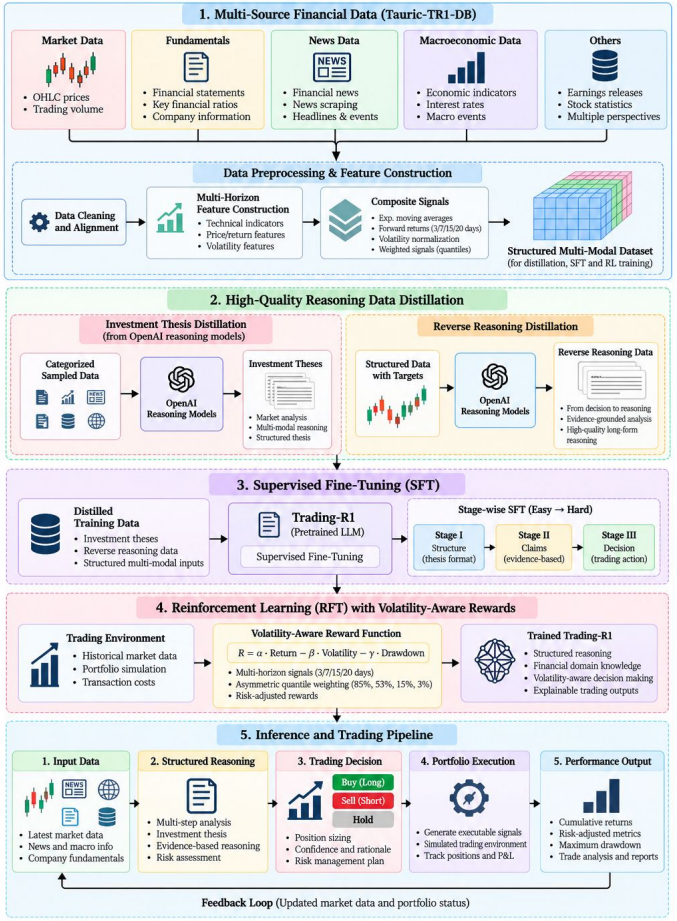
- Modular four-agent LLM architecture.
- Combines technical indicators, chart patterns, trend analysis, and risk management.
- Produces interpretable LONG/SHORT decisions with risk–reward estimates.
- Evaluated on diverse assets using 1-hour and 4-hour market data.
- Reports up to 80% directional accuracy in a continuous SPX case study.

Limitations

- Lower accuracy on 1–15-minute candles.
- LLM and tool-call latency limits real-time HFT suitability.
- Excludes external news and sentiment data.
- Vulnerable to incomplete patterns and premature predictions.
- Live trading performance has not been established by the reported experiments.

Trading-R1: Financial Trading with LLM Reasoning via Reinforcement Learning

Yijia Xio et. al.



Key highlights

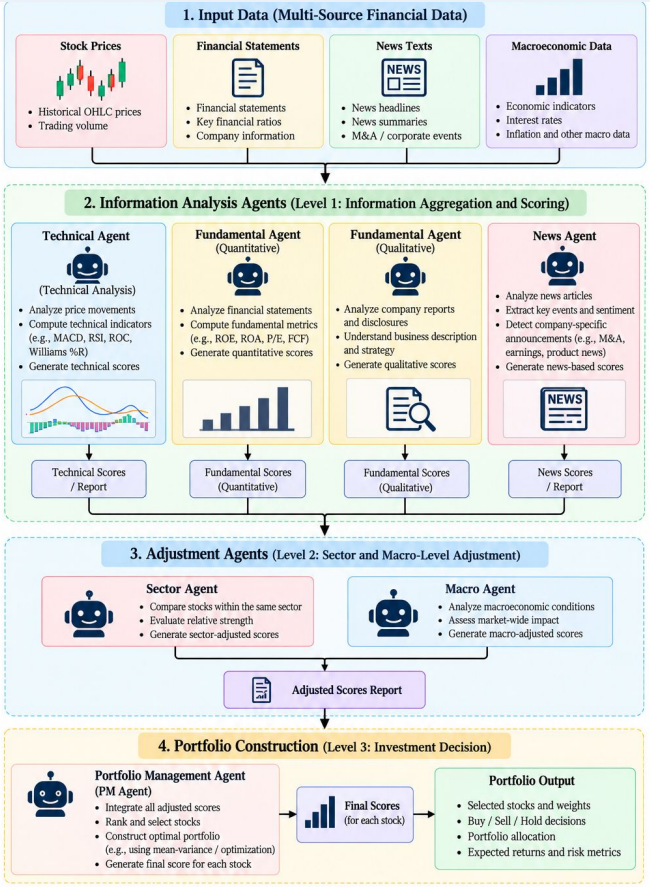
- **Multimodal data**: Integrates market prices, fundamentals, financial news, and macroeconomic indicators.
- **LLM-based reasoning:** Generates structured investment theses and evidence-based decisions.
- **Three-stage training:** Uses SFT for structure and claims, followed by RFT for trading decisions.
- **Volatility-aware rewards**: Considers market volatility and risk-adjusted returns.
- **Explainable decisions**: Produces trading actions — Buy, Sell, or Hold — with supporting reasoning.
- **Portfolio evaluation:** Assesses returns, Sharpe ratio, and maximum drawdown.

Limitations

- **Data dependency**: Performance depends on the quality and availability of financial data.
- **Computational cost:** LLM training and inference can require substantial resources.
- **Market uncertainty:** Sudden market changes may reduce decision accuracy.
- **Reward sensitivity:** Results depend on reward design and risk-related parameters.
- **Execution gap:** Backtesting results may differ from real-world trading due to transaction costs and slippage.
- **Generalization:** Performance across different assets and market conditions requires further validation.

Toward Expert Investment Teams: A Multi-Agent LLM System with Fine-Grained Trading Tasks

Kunihiro Miyazaki et. al.



Key highlights

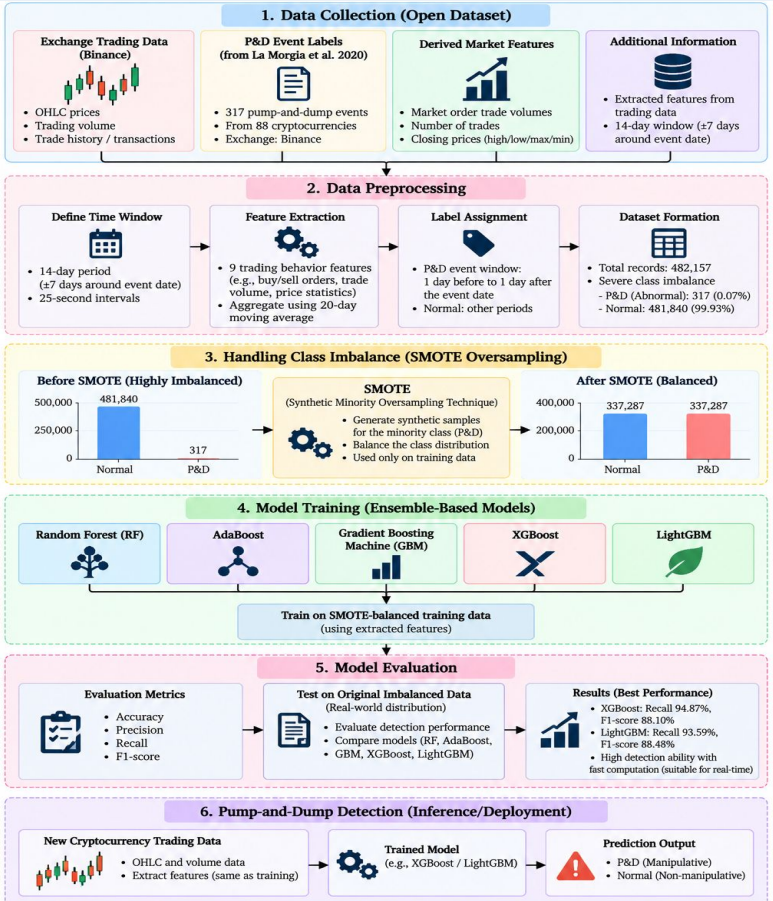
- **Fine-grained analysis:** Improves agent decision-making.
- **Seven specialized agents:** Cover technical, financial, news, sector, and macro analysis.
- **Multi-source data**: Integrates prices, financial statements, news, and economic indicators.
- **Better performance:** Improves Sharpe ratio across 4 of 5 portfolio sizes.
- **Explainability:** Provides scores and reasoning reports.

Limitations

- **Short testing period:** Only 27 months.
- **Limited market scope:** Evaluated on Japanese TOPIX 100 stocks.
- **Agent redundancy:** Some agents introduce noise.
- **Generalization uncertain**: Other markets and LLMs need testing.
- **Live deployment unverified:** Requires further real-world validation.

Improving Cryptocurrency Pump-and-Dump Detection through Ensemble-Based Models and Synthetic Oversampling Techniques

Jieun Yu et. al.



Key highlights

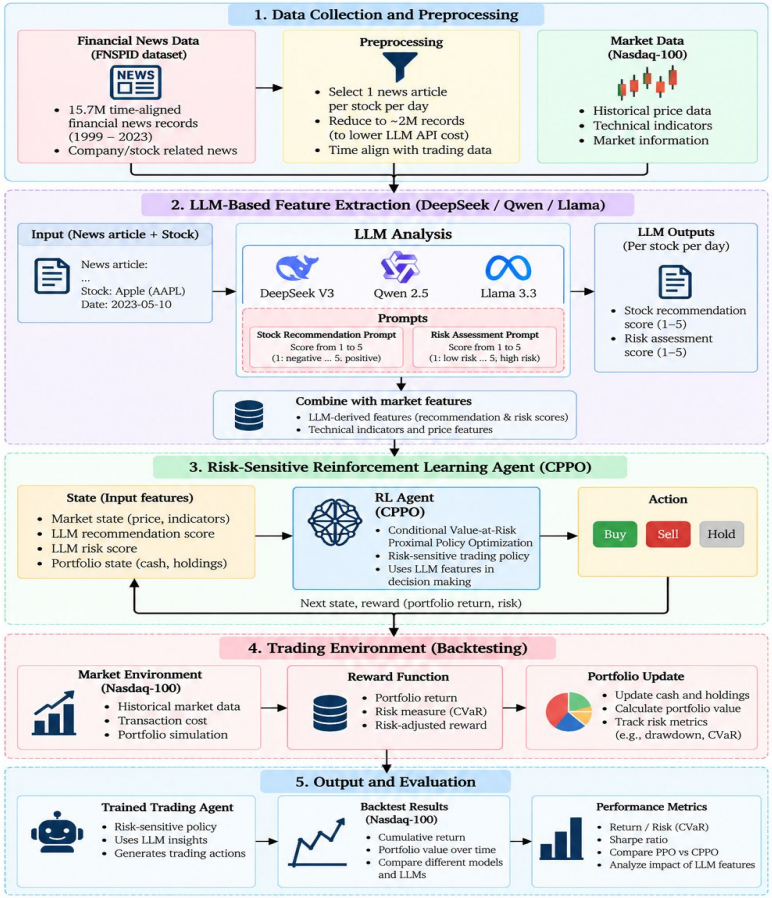
- **SMOTE oversampling**: Addresses severe class imbalance.
- **Five ensemble models**: Compares RF, AdaBoost, GBM, XGBoost, and LightGBM.
- **High recall**: XGBoost achieves 94.87% and LightGBM 93.59%.
- **Fast detection**: XGBoost and LightGBM offer efficient computation.

Limitations

- **Limited dataset**: Uses Binance data only.
- **Feature constraints**: Relies mainly on transaction data.
- **Synthetic noise**: SMOTE may introduce noisy samples.
- **Limited generalization:** Other exchanges and manipulation types need testing.

FinRL-DeepSeek: LLM-Infused Risk-Sensitive Reinforcement Learning for Trading Agents

Mostapha Benhenda



Key highlights

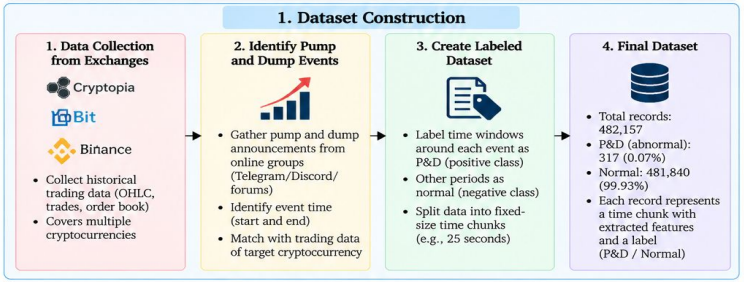
- **LLM integration:** Extracts stock recommendations and risk scores from financial news.
- **Risk-aware trading:** Combines PPO and CVaR-PPO reinforcement learning.
- **Multiple LLMs:** Evaluates DeepSeek V3, Qwen 2.5, and Llama 3.3.
- **Backtesting**: Tests strategies against the Nasdaq-100 benchmark.

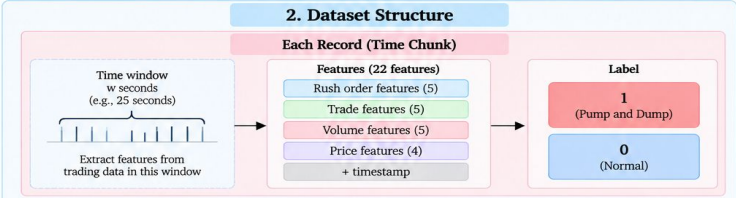
Limitations

- **Inconsistent performance:** LLM integration does not always improve results.
- **High memory usage:** Longer training requires substantial RAM.
- **News quality:** Noisy or weak news signals can affect decisions.
- **Slow adaptation:** Faster market changes require quicker decision-making.

Pump and Dumps in the Bitcoin Era: Real Time Detection of Cryptocurrency Market Manipulations

La Morgia et. al.





Feature Categories and Description

Category	Feature Name	Description
Rush Orders (5)	StdRushOrders	Std. deviation of rush order volume
	AvgRushOrders	Average volume of rush orders
	StdTrades	Std. deviation of number of trades
	StdVolumes	Std. deviation of trade volumes
	AvgVolumes	Average trade volume
Trade Features (5)	StdPrice	Std. deviation of closing prices
	AvgPrice	Average closing price
	AvgPriceMax	Average of highest prices
	AvgPriceMin	Average of lowest prices
	StdTrades	Std. deviation of trades (buy + sell)
Volume Features (5)	AvgVolume	Average trade volume
	StdVolume	Std. deviation of trade volume
	BuyVolume	Total buy volume
	SellVolume	Total sell volume
	VolumeRatio	Buy/Sell volume ratio
Price Features (4)	Open	Opening price (average)
	High	Highest price (average)
	Low	Lowest price (average)
	Close	Closing price (average)
Time	Timestamp	Start time of the chunk (UTC)

Dataset Statistics

Class	Count	Percentage
Normal	481,840	99.93%
Pump & Dump (P&D)	317	0.07%
Total	482,157	100%

Distribution of normal and P&D cases

Dataset Split

Dataset Type	P&D	Normal	Total
Training	222	337,287	144,638 (30%)
Test	95	144,553	337,599 (70%)
Total	317	481,840	482,157

Data split (training vs. test)

Time Chunk Example

Feature	Value (example)	...	Label
StdRushOrders	0.213	...	1
AvgRushOrders	5.432	...	1
StdTrades	2.981	...	1
...	...	...	...
Close	0.00123	...	1

Example of one record in the dataset

Key highlights

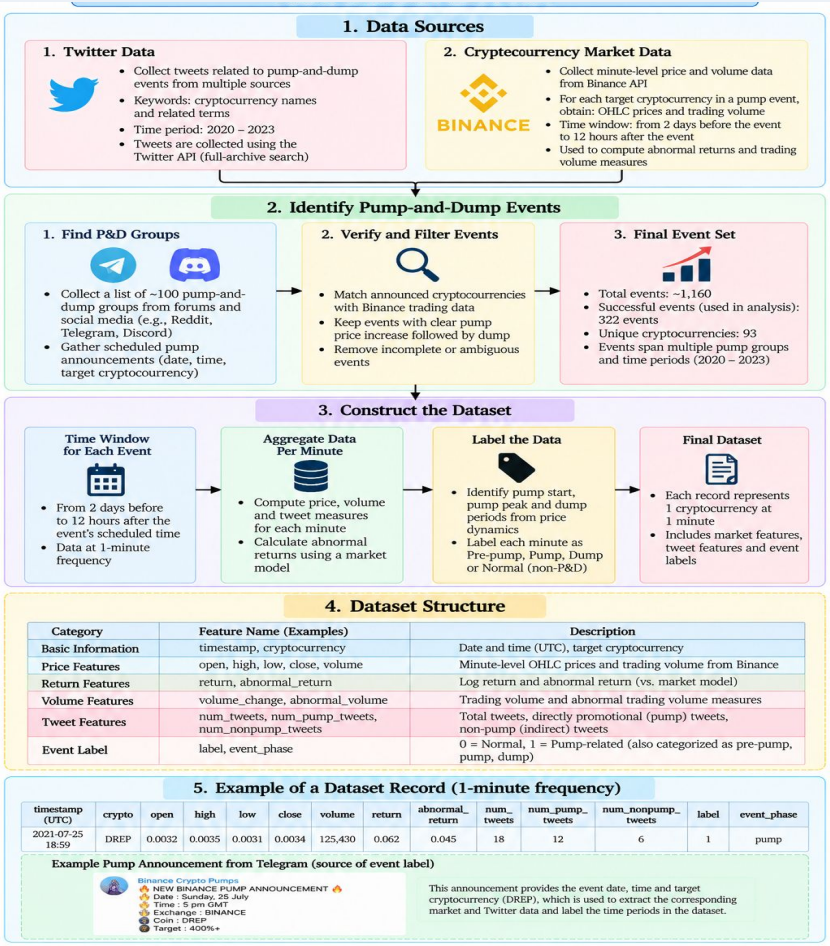
- **Real-time detection:** Detects pump-and-dump schemes within 25 seconds.
- **Rush-order analysis**: Identifies abnormal spikes in market buy orders.
- **Machine learning**: Uses Random Forest and Logistic Regression.
- **High accuracy**: Random Forest achieves an F1-score of 92% with 25-second chunks.
- **Reduced detection time:** Improves on the baseline's 30-minute expected detection time.

Limitations

- **Limited dataset**: Only 104 confirmed Binance events.
- **Incomplete coverage:** Some private or deleted group signals cannot be tracked.
- **Missed events:** Gradual pumping may evade detection.
- **Order inference**: Actual order types are unavailable through the API.
- **False positives:** Unusual legitimate trading activity may trigger alerts.

Twitter and cryptocurrency pump-and-dump

Ardia & Bluteau



Key highlights

- **Large dataset**: 1,160 pump-and-dump events across 93 cryptocurrencies.
- **Multi-source data**: Combines Twitter activity with Binance price and volume data.
- **High-frequency analysis:** Uses one-minute data from 2 days before to 12 hours after each event.
- **Event verification**: Identifies 322 successful pump-and-dump events.
- **Promotion analysis:** Examines how direct and indirect tweets influence market movements.

Limitaions

- **Limited exchange coverage:** Focuses only on Binance and BTC trading pairs.
- **Restricted Twitter data:** API access limitations reduced data availability.
- **Limited tweet coverage:** Excludes some popular cryptocurrencies due to collection constraints.
- **Correlation, not causation:** Cannot conclusively prove Twitter promotion causes price changes.
- **Limited generalization:** Findings may not apply to other exchanges or social platforms.

THANK YOU !